PART ONE: Entities, Attributes, and Relationships

Step 1: List the entities and their attributes

Entity	Attributes	Primary Key
Department	dept_id, dept_name	dept_id
Student	matric_no, student_name, dept_id	matric_no
Lecturer	lecturer_id, lecturer_name, dept_id	lecturer_id
Course	course_code, course_title	course_code
Enrolment	enrollment_id, matric_no, course_code, session	enrollment_id

Step 2: Identify the relationships and their cardinalities

Department to Student: One-to-Many (1 : *) — One department has many students, but each student belongs to only one department.
Student to Course: Many-to-Many (M:N) — One student registers for many courses, and one course has many students.
Course to Lecturer: Many-to-Many (M:N) across sessions — One course can have multiple lecturers across sessions, and one lecturer can teach multiple courses.

Part 2: Normalization from 1NF to 3NF

Step 1: First Normal Form (1NF)

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

This 1NF was derived from breaking cells into smaller records , breaking the list into singular records (codes, course_titles, lecturers, lecturer_depts) into single rows using a composite key of (matric_no, course_code).

Step 2: Second Normal Form (2NF)

Student Table

matric_no	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course Table

course_code	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrollment table

matric_no	course_code	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027

20231235	COS301	2026/2027
20231236	SEN301	2026/2027

I separated the table to remove partial dependencies, ensuring that student_name and student_dept depend solely on matric_no, and course_title depends solely on course_code.

Step 3: Third Normal Form (3NF)

Lecturer Table

lecturer	**dept_name(FK)**
Dr A. Okafor	Software Eng.
Dr M. Bello	Software Eng.
Mrs F. Eze	Computer Sci.

Department Table

dept_name
Software Eng.
Software Eng.
Computer Sci.

LecturerCourse Table

course_code	**course_title**	**session**	**lecturer**
SEN307	Database Systems	2026/2027	Dr A. Okafor
COS301	Operating Systems	2026/2027	Mrs F. Eze
SEN301	Software Architecture	2026/2027	Dr M. Bello

Step 4: Write out your final schema

Department (dept_id PK, dept_name)
Student (matric_no PK, student_name, dept_id FK Department)

Lecturer (lecturer_id PK, lecturer_name, dept_id FK Department)
Course (course_code PK, course_title)
Enrollment (enrollment_id PK, matric_no FK Student, course_code FK Course, session)
LecturerCourse (assignment_id PK, course_code FK Course, lecturer_id FK Lecturer, session)